

# Topology Zoo data-quality audit

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Source: `/home/eric/InternetTopologyZoo/graphml`

Graphs parsed: **276**

Total nodes: **10,952**

Total edges: **13,976**

## Location coverage

- **85** graphs have every node located (lat + lon).
- **175** graphs have a mix of located and unlocated nodes.
- **16** graphs have zero located nodes.
- Total nodes missing at least one of lat/lon: **2,137**

(19.5% of all nodes).

- Nodes whose lat/lon value was *\*provided but invalid\** (empty string, 'None', 'NaN', out-of-range, non-numeric): **0**.

## Graph structure

- Multigraphs (at least one pair of parallel edges): **96**.
- Total parallel edges across the archive: **654**.
- Total self-loops: **2**.
- Graphs with >1 connected component: **23**.
- Isolated (edgeless) nodes overall: **115**.
- Edges whose source or target was not declared as a node: **0**.

## Labels

- Nodes without any `label` attribute: **20**.
- Nodes whose label is whitespace only: **0**.
- Nodes whose label contains control characters: **0**.
- Graphs containing duplicate labels: **120**.
- Total duplicate-label occurrences: **269** (counted as 1 per label that has  $\geq 2$  carriers within its graph).

## Issue tally by category

- node-missing-both: 2137
- edge-parallel: 498
- node-label-duplicate: 266
- node-isolated: 115
- graph-disconnected: 23
- node-label-missing: 20
- edge-self-loop: 2

## Issue tally by severity

- info: 2658
- warn: 403

## Largest graphs

| Graph         | Nodes | Edges | Multi? | Missing lation |
|---------------|-------|-------|--------|----------------|
| Kdl           | 754   | 899   | yes    | 28             |
| Cogentco      | 197   | 245   | yes    | 11             |
| DialtelecomCz | 193   | 151   | no     | 15             |
| UsCarrier     | 158   | 189   | no     | 6              |
| Colt          | 153   | 191   | yes    | 4              |
| GtsCe         | 149   | 193   | no     | 8              |
| TataNld       | 145   | 194   | yes    | 2              |
| Pern          | 127   | 129   | no     | 119            |
| Ion           | 125   | 150   | yes    | 22             |
| Deltacom      | 113   | 183   | yes    | 12             |

## Worst location coverage (absolute count)

| Graph         | Nodes | Missing | %      |
|---------------|-------|---------|--------|
| Pern          | 127   | 119     | 93.7%  |
| Opteglobe     | 93    | 93      | 100.0% |
| IntNetworkmap | 89    | 89      | 100.0% |
| AsnetAm       | 65    | 64      | 98.5%  |
| Garr          | 56    | 56      | 100.0% |
| Cudi          | 51    | 51      | 100.0% |
| Esnet         | 68    | 51      | 75.0%  |

| Graph     | Nodes | Missing | %      |
|-----------|-------|---------|--------|
| Internode | 66    | 46      | 69.7%  |
| Columbus  | 70    | 39      | 55.7%  |
| Geant     | 37    | 37      | 100.0% |

## Top multigraphs by parallel-edge count

| Graph      | Edges | Parallel pairs |
|------------|-------|----------------|
| Ntt        | 216   | 153            |
| Deltacom   | 183   | 22             |
| Highwinds  | 53    | 22             |
| Globenet   | 113   | 18             |
| Sunet      | 49    | 17             |
| Colt       | 191   | 14             |
| Garr201112 | 89    | 14             |
| Garr201201 | 89    | 14             |
| Esnet      | 92    | 13             |
| Garr201110 | 87    | 13             |

## How to use the workbook

The companion workbook `zoo-audit.xlsx` has:

1. **Summary** — one row per graph with the same columns as the totals at the bottom of this report. The last row is the cross-archive total.
2. **Missing Locations** — every single node with a missing or invalid lat/lon across the whole corpus.
3. **Issues** — every issue found, labelled by severity (error / warn / info) and category, with the node or edge id in the detail.
4. **Multigraphs** — the subset of graphs with parallel edges, plus their edge counts.
5. **one sheet per graph** — full attribute table for every node and every edge; rows with problems are highlighted.

## Notes on terminology

- "Missing lat/lon" counts a node if *either* of its Latitude or Longitude attributes is absent, empty, one of the sentinels `None/NaN/null/-`, or fails numeric coercion.
- "Multigraph" here means *observed* parallel edges, not the GraphML `parse.multiplegraph` flag (which is rarely set in the Zoo files).
- "Duplicate labels" counts **distinct labels** carried by 2+ nodes; for a count of *affected nodes* multiply by at least 2.